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From proficiency to pedagogy: A mixed-methods study of in-service teachers' TPACK-GenAI and the mediating role of pedagogical knowledge

Laila Mohebi [a,*], Areej ElSayary [b]

[a] College of Humanities and Social Sciences, Zayed University, United Arab Emirates
[b] College of Interdisciplinary Studies, Zayed University, United Arab Emirates

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ABSTRACT

This study investigated in-service teachers' Technological Pedagogical Content Knowledge for Generative Artificial Intelligence (TPACK-GenAI) and examined the factors influencing their integration of generative AI in classroom practice. Using an explanatory sequential mixed-methods design, data were collected from 325 in-service teachers across 26 countries, followed by in-depth interviews with seven teachers. Quantitative results revealed that Technological Knowledge (TK), Pedagogical Knowledge (PK), and Pedagogical Content Knowledge (PCK) were significantly associated with teachers' overall TPACK-GenAI. Crucially, mediation analysis confirmed that Technological Pedagogical Knowledge (TPK) serves as the indirect association through which TK is related to overall TPACK-GenAI competence. Qualitative findings enriched this result, illustrating that teachers' confidence stems not from technical skill alone, but from their ability to apply AI tools to specific pedagogical challenges. The qualitative data also provided context for the non-significant effects of demographic variables, suggesting that barriers such as lack of access and training are more influential than experience or school level. These findings emphasize the need for professional development that is pedagogically focused, context-sensitive, and moves beyond mere technical training. This study contributes a detailed, multinational perspective on teacher preparedness in the age of AI and validates the critical role of pedagogical thinking in effective technology integration.

1. Introduction

In today's digital era, Generative AI (GenAI) tools and technologies offer immense potential to transform education (Guler et al., 2024). These innovations can help teachers to diagnose specific learning challenges and deliver tailored interventions, making learning more efficient and effective (Li et al., 2023). For instance, Wei (2023) revealed that GenAI tools can help educators identify where students are struggling and provide tailored support to address those difficulties (Wang & Wang, 2022). Research has also shown that GenAI tools positively influence how students learn and how they feel about using technology (Nazari et al., 2021; Utami et al., 2023).

Regardless of these promising benefits, scholars have raised valid concerns about the accuracy of AI-generated content in certain subject areas and the risk of students becoming too dependent on these tools (Matzakos et al., 2023). This reflects that GenAI should be seen as a helpful supplement that enhances teaching rather than something that replaces traditional instruction. Successfully integrating GenAI into classroom practice requires teachers to have more than just technical skills; they need to understand how to effectively blend this technology with their teaching strategies and subject knowledge, which is captured in the TPACK (Technological Pedagogical Content Knowledge) framework. When in-service teachers lack confidence in their TPACK-GenAI abilities, they often struggle to effectively incorporate GenAI technologies into their lessons (Dasari et al., 2024). This emphasizes why it is so imperative to examine how in-service teachers view their TPACK-GenAI confidence and in what way their perceptions affect their classroom use of GenAI tools. Understanding teachers' confidence levels and integration practices is crucial for identifying what is holding them back and for creating professional development programs that truly address their needs (Getenet, 2024). Despite the growing interest in TPACK frameworks, studies focusing specifically on in-service teachers' TPACK in GenAI contexts remain limited, highlighting a gap in empirical research (Chai et al., 2013; Voogt et al., 2015). Most existing literature emphasizes pre-service teacher training or general TPACK development, indicating that the unique challenges and professional growth of in-service

* Corresponding author.
E-mail addresses: laila.mohebi@zu.ac.ae (L. Mohebi), areej.elsayary@zu.ac.ae (A. ElSayary).

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teachers in integrating technology are still underexplored (Angeli & Valanides, 2009). However, there is still inadequate research examining in-service teachers' TPACK-GenAI confidence and how they are using AI tools in real classroom settings (Yadav et al., 2026). While existing studies highlight the importance of professional development in helping teachers adopt new technologies (Daher et al., 2021), there is a clear need for research that explores the connection between teachers' TPACK-GenAI confidence and their actual implementation of GenAI in their classrooms.

1.1. Purpose of the study

In an era of rapid digital transformation, the integration of GenAI into education has become a global priority. Nations worldwide are developing strategies to leverage AI for educational innovation, reflecting a shared ambition to prepare students for a technology-driven future (Tu & Lu, 2025). While specific national initiatives, such as the United Arab Emirates' (UAE) vision for smart education, provide illustrative examples of this trend, the challenges and opportunities for teacher preparedness are transnational. The successful integration of GenAI in classroom practice relies not only on infrastructure but also on in-service teachers' confidence and competence in applying TPACK within this new domain. Discovering teachers' TPACK-GenAI confidence is therefore necessary for understanding in what way educators in the UAE navigate the opportunities and challenges of GenAI adoption. This study, therefore, adopts a multinational perspective to investigate how in-service teachers across diverse educational contexts perceive their readiness to integrate GenAI. The sample, comprising educators from 26 countries, allows for an exploration of the common factors that shape teachers' TPACK-GenAI competencies, contributing to a broader, more generalizable understanding of this global phenomenon.

The following research questions were formulated to guide this study:

1. What are the in-service teachers' perceptions about their confidence in using TPACK-GenAI competencies?
2. What is the relationship between the teachers' TPACK-GenAI domains?
3. To what extent do teaching experience and school level relate to teachers' TPACK-GenAI competencies?
4. Does Technological Pedagogical Knowledge (TPK) mediate the relationship between Technology Knowledge (TK) and TPACK?
5. Does school level moderate the relationship between TK and TPACK?
6. What challenges and supports do in-service teachers report when using GenAI in their teaching?

The following hypotheses were made:

H1. There are significant positive correlations between the TPACK-GenAI domains.

H2. Teaching experience and school level are significantly associated with teachers' self-reported TPACK-GenAI competencies.

H3. TPK significantly mediates the relationship between TK and TPACK.

H4. School level significantly moderates the relationship between TK and TPACK, such that the relationship varies across different school levels (e.g., kindergarten, primary, secondary).

2. Literature review

2.1. Conceptualizing Technological Pedagogical Content Knowledge (TPACK)

The TPACK framework, fundamentally conceptualized by Mishra and Koehler (2006), offers a comprehensive lens for understanding and promoting the integration of technology into teaching. TPACK emphasizes the vibrant interplay between three core domains of teacher knowledge: TK, pedagogical knowledge (PK), and content knowledge (CK). Effective technology-enhanced instruction depends not only on teachers' competency in each domain independently but also on their ability to integrate them implicitly in instructional design and delivery. As a widely recognized model, TPACK has been comprehensively applied in teacher education to assess and support both pre-service and in-service teachers in developing the competencies required for integrating digital tools into pedagogical practices (Mayer & Girwidz, 2019). Further, recently, Mishra et al. (2023) reconceptualized the framework in the context of GenAI, emphasizing that the generative and interactive affordances of these tools claim a critical reassessment of pedagogical approaches, assessment strategies, and disciplinary knowledge. Building on this perception, the current study situates its investigation within the TPACK framework to explore how teachers navigate the intersections of content, pedagogy, and emerging GenAI technologies (Lakhe Shrestha et al., 2025).

2.2. TPACK framework with GenAI for professional development

The TPACK framework highlights the integration of technology into teaching by combining TK, PK, and CK to acquire effective educational practices. In the context of this study, the framework offers a groundwork for understanding in what way GenAI (TK) can be profoundly integrated with teaching strategies (PK) to enhance instructional practices and help professional development (Mishra & Koehler, 2006). By ensuring a composed integration of these domains, the TPACK framework operates as a guide for leveraging GenAI tools in teaching, while also adopting the challenges and realistic strategies associated with their application (Chai et al., 2013). Ning et al. (2024) further highlighted that the benefits of GenAI can only be amplified when digital tools are aligned with pedagogical goals and disciplinary content. Hence, the TPACK framework provides a critical lens for observing how GenAI can contribute to teachers' confidence to promote effective classroom practices (Tbaishat et al., 2025).

2.3. Positive correlations among TPACK-GenAI domains

Several studies have consistently reported robust positive correlations among the components of TPACK, including AI-specific extensions. For instance, Karatas and Ataç (2025) examined English pre-service teachers in Turkey and found that while participants were proficient in TK and PK, they struggled to integrate these domains and demonstrated limited understanding of AI ethics. Their findings highlight the need for targeted teacher education programs that build integrated and ethical AI-pedagogical competencies. Likewise, in a study conducted in China, Ning et al. (2024) reported strong correlations between AI-related knowledge components such as AI-Technological Knowledge (AI-TK), AI-Technological Content Knowledge (AI-TCK), and AI-Technological Pedagogical Knowledge (AI-TPK) and overall AI-TPACK, whereas CK and PK exhibited weaker associations. Sofwan, Habibi, and Yaakob (2023) extended these findings in a Malaysian pre-service teacher sample, observing that TPK exhibited the strongest relationship with overall TPACK. Collectively, these studies endorse that the various TPACK-GenAI domains co-vary positively, as teachers who demonstrate strength in one knowledge domain tend to exhibit higher competence across others.

2.4. Teachers' characteristics as predictors of TPACK-GenAI competencies

Research evidence suggests that teacher characteristics and professional development are significant predictors of self-reported AI-TPACK competencies. Supporting this, Hava and Babayigit (2025) examined 401 Turkish teachers and found that higher levels of digital proficiency as a proxy for professional development were associated with significantly stronger AI-TPACK skills. Similarly, Prasetya et al. (2024) reported that teachers with more than ten years of experience demonstrated more positive attitudes toward AI integration and higher AI-related competencies compared to those with less than five years of experience. In addition, Shrestha et al. (2025) highlighted that structured institutional training combined with peer collaboration fosters the development of GenAI-related competencies aligned with TPACK principles. Their findings further emphasized that targeted training programs enhance teachers' preparedness and attitudes toward integrating AI in educational settings. Collectively, these studies indicate that professional development and teaching experience play a crucial role in predicting teachers' TPACK-GenAI competencies.

2.5. Mediating and moderating mechanisms within the TPACK–GenAI framework

Research evidence indicates that TPK serves as a crucial mechanism through which foundational TK translates into the actual classroom use of AI tools. For example, Sofwan, Habibi, and Yaakob (2023) examined technology integration models among Malaysian pre-service teachers and found that TPK was the strongest predictor of technology integration, functioning as a mediating component within the broader TPACK framework. Their structural equation modeling demonstrated substantial indirect effects of TPK on technology integration through overall TPACK. Similarly, Al-Abdullatif (2024) investigated GenAI adoption among teachers and confirmed that the *Intelligent-TPACK* construct mediated the relationship between AI literacy (analogous to TK) and GenAI acceptance. Collectively, these findings suggest that higher levels of PCK bridge the gap between basic technological skills and their effective instructional application. Conversely, limited research has directly examined school level as a moderating factor in the relationship between TPACK competencies and GenAI use (Liu & Zhong, 2025). Although contextual factors may influence the strength of the TPACK use relationship, evidence from a meta-analysis of general TPACK studies indicates that the teaching stage (primary versus secondary) does not significantly moderate the correlation between technology self-efficacy and TPACK (Backfisch et al., 2025). Thus, while variations in AI integration across educational contexts are evident, empirical support for school level as a moderator of the TPACK-GenAI competencies-use relationship remains inconclusive. The integration of GenAI in education is particularly critical in the context of the UAE, where higher education institutions are at the forefront of digital transformation. The UAE has invested heavily in technological innovation to advance its national vision of becoming a global leader in AI and smart education.

3. Methodology

This study employed an explanatory sequential mixed-methods design, combining quantitative and qualitative approaches to provide a comprehensive understanding of in-service teachers' TPACK-GenAI competencies and their integration of GenAI in classroom practice. In this design, the quantitative phase was conducted first to test hypothesized relationships among variables, followed by a qualitative phase that further explained and contextualized the statistical findings. The survey was distributed online using a secure survey platform (e.g., Qualtrics/Google Forms), and invitations were sent via email to in-service teachers through professional networks and school contacts. Prior to participation, all respondents were provided with an electronic written consent form detailing the purpose of the study, confidentiality assurances, and their right to withdraw at any time. Consent was obtained electronically before participants could access the survey.

3.1. Sample

Table 1 presents the demographic and professional background of the study participants. The sample consisted of 325 participants, 63.7% of whom were female and 36.3% were male. The majority of participants fell within the 25-30 age group (27.7%), followed by those aged 30-35 years (23.7%), 35-45 years (20.9%), and over 45 years (16.3%). A smaller proportion (11.4%) was under 25 years of age. Regarding academic qualifications, over half of the participants held a bachelor's degree (52.9%), followed by those with a master's degree (31.4%), a diploma (8.3%), and a PhD (7.4%). Participants taught across various domains, with the highest representation in scientific subjects (40.9%), followed by literary subjects (34.5%), early childhood education (ECE) (15.4%), arts education (5.2%), and special education (4.0%).

A substantial majority (84.3%) reported working as teachers at the time of the study. Teaching experience varied, with 33.5% having 1-5 years of experience, 19.1% with 6-10 years, and 15.4% with 11-15 years. Participants with less than 1 year of experience (12.0%), 16-20 years (8.9%), and more than 20 years (11.1%) were also represented. Regarding school level, most participants taught at the high school level (37.8%), followed by primary (25.5%), middle (23.7%), and kindergarten (12.9%). Notably, 83.1% of the participants reported that they had experience teaching AI, while 16.9% did not. This background information emphasizes a diverse and experienced sample, relevant for exploring issues related to AI in education.

Fig. 1 illustrates the distribution of participants by nationality. The countries were selected based on diversity in educational systems, technological integration in classrooms, and availability of willing teacher participants. This selection allowed for comparative insights

Table 1
Summary of the participants' background information.

Variables	F	Percentage (%)
Gender		
Male	118	36.3
Female	207	63.7
Age		
≤25 years	37	11.4
25-30 years	90	27.7
30-35 years	77	23.7
35-45 years	68	20.9
≥45 years	53	16.3
Qualification		
Diploma	27	8.3
Bachelor's	172	52.9
Master's	102	31.4
PhD	24	7.4
Area of Teaching		
Scientific subject	133	40.9
Literary subjects	112	34.5
ECE	50	15.4
Special education	13	4.0
Arts education	17	5.2
Working as a Teacher		
Yes	274	84.3
No	51	15.7
Teaching Experience		
≤1 year	39	12.0
1-5 years	109	33.5
6-10 years	62	19.1
11-15 years	50	15.4
16-20 years	29	8.9
≥20 years	36	11.1
Nature of School		
Kindergarten	42	12.9
Primary school	83	25.5
Middle school	77	23.7
High school	123	37.8
Teaching of AI		
Yes	270	83.1
No	55	16.9

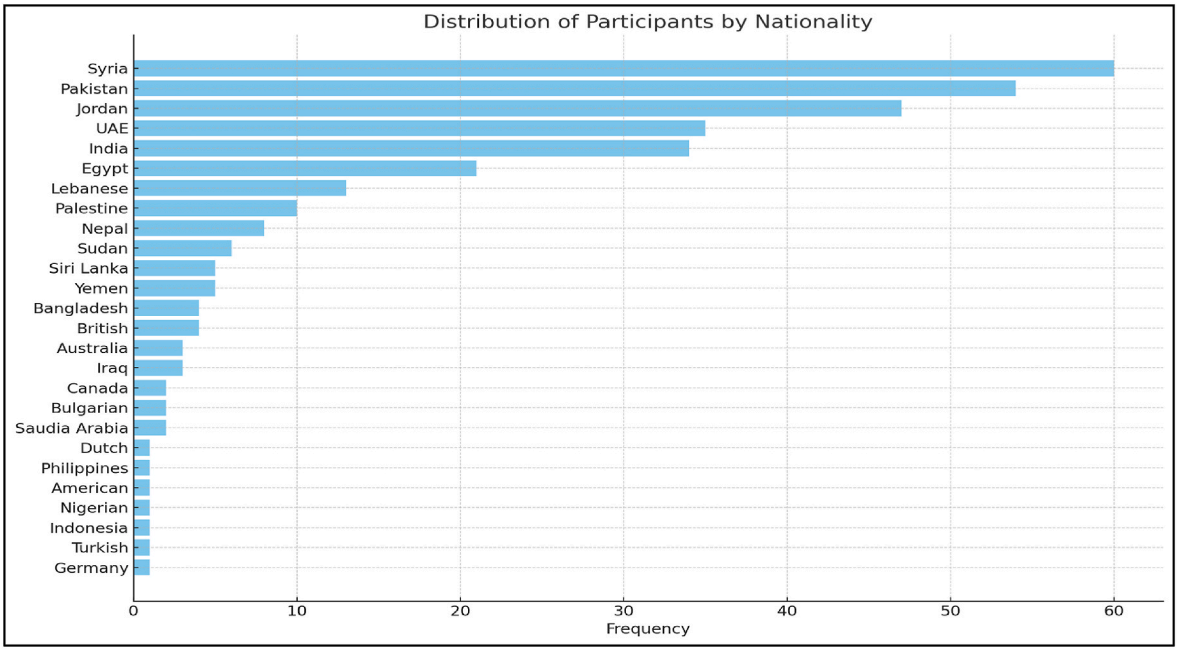

Figure-1. Distribution of participants by nationality.

into how TPACK-GenAI competencies vary across different educational contexts.

The sample comprised individuals from 26 countries, with the majority representing Syria (18.5%), Pakistan (16.6%), and Jordan (14.5%). These three nationalities alone accounted for nearly half of the total participants. Other countries with notable representation included the UAE (10.8%), India (10.5%), and Egypt (6.5%). Several nationalities, such as Dutch, American, Nigerian, Turkish, Indonesian, and German, were represented by only a single participant each (.3%). The overall distribution indicates a diverse, yet unevenly represented, international sample, with a stronger presence from Middle Eastern and South Asian countries (see Fig. 1).

3.2. Measure

The TPACK-GenAI scale was developed by the authors based on a thorough review of existing literature on TPACK and recent studies on AI integration in education. Items were adapted and refined to reflect teachers' knowledge, confidence, and practical use of GenAI tools in classroom settings. Items for the core TPACK constructs were adapted from the widely validated survey by Schmid et al. (2020), while the GenAI-specific items were newly developed and refined for this study based on the conceptual work of Mishra et al. (2023) and the AI literacy framework proposed by Ng et al. (2023). A pilot study was conducted to ensure the reliability and validity of the instrument. The full survey instrument is included in the Appendix for reference. After receiving the invitation to participate, respondents completed an online survey. The survey began with demographic questions, including age, gender, nationality, educational institution, and the frequency of AI use in daily activities. The second section of the survey was organized around the study's core variables and divided into three main categories, each measuring different aspects of teachers' TPACK-GenAI competencies and their classroom integration. All surveys and interviews were conducted in English across all participating countries. This decision was based on the fact that English is widely used in the educational institutions included in the study, and all participants demonstrated sufficient proficiency to understand and respond to survey items and interview questions. Where necessary, clarifications were provided by the research team.

3.2.1. TPACK-GenAI scale

The overall internal consistency of the TPACK-GenAI scale, which consists of 43 items, was assessed using Cronbach's alpha. The result revealed an excellent reliability coefficient (α = .979), indicating that the scale has very high internal consistency. This suggests that the items consistently measure the intended construct and that the scale is highly reliable for assessing TPACK-GenAI competencies. To examine the internal consistency of each TPACK-GenAI domain, Cronbach's alpha coefficients were calculated. The results indicate high reliability across all subscales. All alpha values exceed the commonly accepted threshold of .70 (Nunnally, 1978), demonstrating that each scale possesses acceptable to excellent internal consistency. The TPK subscale showed the highest reliability (α = .952), while CK, with the fewest items, still demonstrated strong reliability (α = .853). These results confirm that the instruments used for measuring TPACK-GenAI domains are statistically reliable and appropriate for further analysis (see Table 2).

3.3. Reliability and validity

Upon receiving ethical approval (Approval No. XXXXXX), written informed consent was obtained from all participants. A preliminary pilot study was conducted to refine the questionnaire, ensuring both its reliability and validity in measuring the constructs of interest. Exploratory factor analysis (EFA) and reliability testing (Cronbach's alpha) were employed to confirm the internal consistency and construct validity of the instrument (Hair et al., 2019). An EFA using principal component

Table 2
Internal consistency of TPACK-GenAI subscales.

Subscales	Items	Cronbach's Alpha
Technological Knowledge	6	.905
Content Knowledge	3	.853
Pedagogical Knowledge	6	.906
Pedagogical Content Knowledge	5	.899
Technological Content Knowledge	6	.921
Technological Pedagogical Knowledge	9	.952
Technological Pedagogy and Content Knowledge	4	.896
Technological Pedagogy and Content Knowledge Competencies	4	.880

analysis with Varimax rotation was conducted on 43 items to identify the underlying factor structure. The Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy was .966, exceeding the recommended minimum value of .60 (Kaiser, 1974), and indicating that the sample size was more than adequate. Bartlett's test of sphericity, χ^2 (903) = 13236.916, $p <$ .001, was statistically significant, suggesting that the correlation matrix was not an identity matrix and that the variables were sufficiently interrelated for factor analysis. Four components with eigenvalues greater than 1.0 were extracted, explaining 68.24% of the total variance. Component loadings ranged from .55 to .83, with communalities between .52 and .82, indicating that the retained factors accounted for a substantial proportion of variance in each item. The rotated factor solution revealed a clear and interpretable structure, with items loading strongly on their respective components and minimal cross-loadings, supporting the multidimensional nature of the construct. Further, confirmatory factor analysis (CFA) was conducted to validate the four-factor structure identified in the EFA. Factor loadings ranged from .65 to .82, exceeding the commonly recommended threshold of .50 (Hair et al., 2019), and all were statistically significant, demonstrating strong indicator reliability. Communalities ranged from .65 to .82, indicating that a substantial proportion of variance in each indicator was explained by its corresponding latent factor. The four-factor structure maps onto the eight domain scores as follows: Factor 1 encompasses the core technological domains (TK, TCK), Factor 2 represents the pedagogical domains (PK, PCK), Factor 3 captures content knowledge (CK), and Factor 4 represents the integrated domains (TPK, TPACK, TPACKCOM). Domain scores were computed as the mean of items loading on each respective subscale within these factors. The model fit indices met acceptable thresholds (e.g., CFI >.90, RMSEA <.08, SRMR <.08), suggesting a good fit between the hypothesized model and the observed data. These results provide empirical support for the construct validity of the measurement model. Following the pilot study, the refined survey was disseminated to the target group after participants were informed of the study's aims and procedures.

3.4. Semi-structured interview

Semi-structured interviews were conducted to gain an in-depth understanding of in-service teachers' perceptions, experiences, and confidence regarding the integration of GenAI within their classroom practices, framed through the TPACK-GenAI model. A purposive sample of twelve in-service teachers who had previously attended professional development training on GenAI tools was invited to participate. Of these, seven teachers consented and were available for interviews, resulting in a final qualitative sample of (n = 7). The interviews aimed to explore teachers' self-perceived competencies across the TPACK-GenAI domains TK, PK, CK, and their intersections (TPK, TCK, PCK, and TPACK) as well as their approaches to integrating GenAI into teaching, learning, and assessment processes. Each interview lasted approximately 40-60 min (average duration was 50 min) and was conducted online via the Zoom platform. To facilitate reflective and meaningful responses, open-ended questions were developed using the Rolfe et al. (2001) reflective model, structured around the prompts *"What?"*, *"So what?"*, and *"Now what?"* These reflective questions encouraged participants to describe their GenAI experiences, interpret their significance, and articulate future applications or needs for professional growth. To ensure content and face validity, the interview protocol was reviewed by two experts in educational technology and teacher professional development. The experts evaluated question clarity, alignment with the study objectives, and relevance to the TPACK-GenAI framework. Based on their feedback, repetitive items were eliminated, and question wording was refined for precision and coherence. Examples of interview questions are given below:

1. What type of AI do you believe has the greatest potential to revolutionize education, and why?
2. Can you describe an innovative use of AI in education that you found particularly effective, and explain what made it stand out?
3. Please share a specific example of how you have used AI to enhance student learning and engagement in your classroom. What factors contributed to its success?
4. What challenges do you face in integrating AI tools into your teaching practice, and what kinds of institutional support or resources would help you use AI more effectively?

3.5. Procedure

This study began with quantitative survey data collection, followed by qualitative interviews. The quantitative phase examined in-service teachers' TPACK-GenAI confidence and their integration of GenAI tools across three domains: *AI for content creation and instructional* support, *AI for teaching and learning*, and *AI for assessment and feedback*. An anonymous online survey was distributed after obtaining informed consent. Based on the quantitative results, seven in-service teachers were purposively selected for semi-structured interviews to gain deeper insights into their experiences and challenges. The interviews, lasting about 50 min each, were conducted via Zoom using reflective, open-ended questions. Qualitative data was analyzed using both deductive and inductive coding. Deductive coding followed pre-defined categories aligned with the study domains, while inductive coding identified emerging themes. To ensure the trustworthiness of the qualitative data, several established procedures were followed (Guba & Lincoln, 1983). First, to enhance credibility, member checking was conducted with three of the seven participants, who reviewed the summary of their interview transcripts and the initial thematic analysis to verify that their perspectives were accurately represented (McKim, 2023). Second, for dependability and confirmability, two of the researchers independently coded the interview transcripts and then met to discuss and resolve any discrepancies, achieving an inter-coder agreement of 92%. An audit trail, consisting of raw data, data reduction and analysis notes, and data reconstruction and synthesis products, was maintained throughout the research process (Lan et al., 2025).

3.6. Data analysis

To address the first research question, Pearson's correlation test was conducted to examine the relationships among teachers' TPACK-GenAI domains. A multiple regression analysis was then performed to assess whether professional development opportunities, teaching experience, and school level significantly predicted teachers' TPACK-GenAI competencies. Furthermore, mediation analysis was conducted using the PROCESS macro (Hayes, 2017) to test whether TPK mediated the relationship between TK and overall TPACK-GenAI. In addition, moderation analysis was employed to examine whether school level moderated the relationship between TK and TPACK, such that the strength of this relationship varied across school contexts.

4. Results

4.1. Relationship between the teachers' TPACK-GenAI domains

Table 3 presents the detailed values of descriptive statistics and correlation analysis for the TPACK framework components examined in the study. Fig. 2 has been included to visually illustrate the average scores of each TPACK construct such as TK, CK, PK, and their integrated forms (PCK, TCK, TPK, TPACK, and technological pedagogy and content knowledge competencies [TPACKCOM]) to allow a clear comparison of construct prominence among participants. These constructs represent teachers' self-assessed competencies across different domains of technological, pedagogical, and content knowledge integration. The correlation matrix demonstrates significant positive relationships among all variables, suggesting that as one area of knowledge increases, related

Table 3
Descriptive statistics and correlation among the teachers' TPACK-GenAI domains.

	Variables	*M*	*SD*	*1*	*2*	*3*	*4*	*5*	*6*	*7*
1.	TK	24.46	4.512							
2.	CK	12.84	2.21	.680[a]						
3.	PK	25.59	4.12	.645[a]	.819[a]					
4.	PCK	20.93	3.29	629[a]	.738[a]	.825[a]				
5.	TCK	24.75	4.40	701[a]	.573[a]	.640[a]	.704[a]			
6.	TPK	37.02	6.60	.744[a]	.595[a]	.656[a]	.687[a]	.881[a]		
7.	TPACK	16.42	2.92	.683[a]	.579[a]	.650[a]	.673[a]	.784[a]	.865[a]	
8.	TPACKCOM	16.47	3.00	.664[a]	.536[a]	.599[a]	.622[a]	.763[a]	.805[a]	.841[a]

Note. *TK* = technological knowledge; *CL* = content knowledge; *PK* = pedagogical knowledge; *PCK* = pedagogical content knowledge; *TCK* = technological content knowledge; *TPK* = technological pedagogical knowledge; technological pedagogy and content knowledge; TPACK = technological pedagogy and content knowledge; *TPACKCOM* = technological pedagogy and content knowledge competencies.
[a] Correlation is significant at the .01 level (2-tailed).

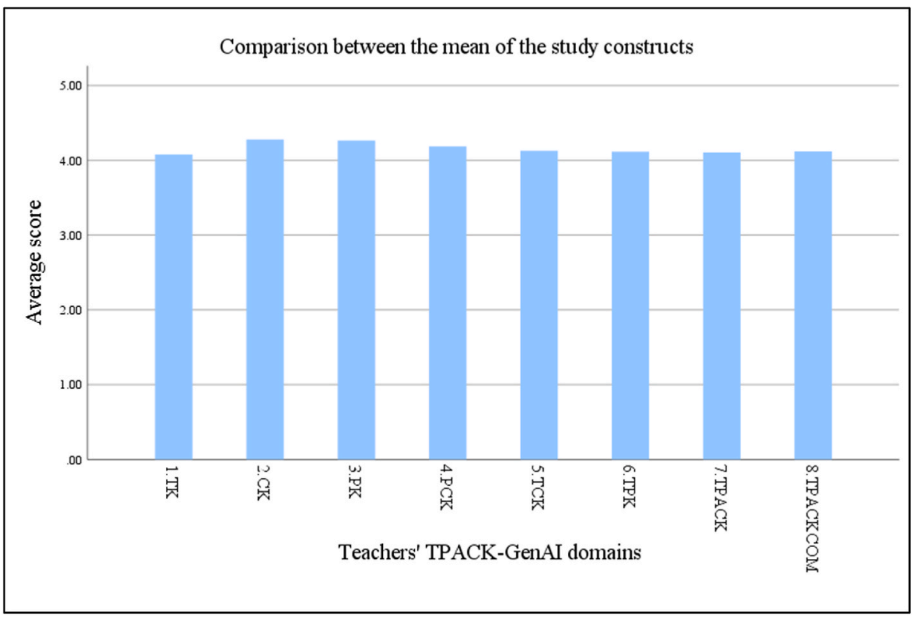

Fig. 2. The average scores of teachers' TPACK-GenAI domains.

competencies also tend to strengthen. This highlights the interconnected nature of the TPACK components in effective teaching practices (see Fig. 2) (see Table 4).

Table 3 shows the mean values, standard deviations, and Pearson correlation coefficients among the core domains of teachers' TK and TPACK components. All correlations were statistically significant at $p < .01$, indicating strong and meaningful relationships between variables. TK had strong positive correlations with TCK (r = .701), TPK (r = .744), and the overall TPACK score (r = .683). This suggests that as teachers' technological knowledge increases, their ability to integrate technology with pedagogy and content also strengthens. CK showed high correlations with PK (r = .819) and PCK (r = .738), indicating that teachers with strong CK are more likely to demonstrate sound pedagogical understanding and integration. PK and PCK also had a high intercorrelation (r = .825), highlighting the close relationship between general pedagogical skills and their application within subject content areas. TCK, TPK, and PCK all had strong correlations with TPACK (TCK-TPACK: r = .784; TPK-TPACK: r = .865; PCK-TPACK: r = .673), reinforcing that the integrated knowledge of technology, pedagogy, and content is most effective when each domain is well-developed. The highest correlation observed was between TPK and TPACK (r = .865), suggesting that the ability to use technology in pedagogically sound ways plays a critical role in achieving overall TPACK competence. Lastly, TPACKCOM also exhibit strong correlations, particularly with TPACK (r = .841), validating the alignment between theoretical knowledge and applied competency. These results confirm that the TPACK-GenAI domains are conceptually and statistically interlinked, providing empirical support for the validity of the TPACK model in the context of AI-integrated teaching. Therefore, hypothesis 1 is accepted.

4.2. Multiple regression analysis

Multiple linear regression was conducted to examine the extent to which the predictor variables explain variance in the outcome variable. This method allows for assessing the unique contribution of each predictor while controlling for the effects of others. The analysis generated unstandardized B-coefficients, which indicate the expected change in the dependent variable for each unit change in the predictor, holding other variables constant. To visually interpret the relative impact of each predictor, a graph was developed plotting the B-coefficients on the Y-axis against the corresponding predictors on the X-axis, providing a clear comparison of their effects (see Fig. 3).

As shown in Table 4, a multiple regression analysis was conducted to examine whether TK, PK, CK, PCK, teaching experience, level of school, and gender predicted TPACK scores. The overall model was statistically significant, $F(7, 317) = 61.35$, $p < .001$, explaining approximately 57.5% of the variance in TPACK scores ($R^2 = .58$, adjusted $R^2 = .57$).

Among the predictors, TK ($B = .28$, $\beta = .43$, $p < .001$), PK ($B = .15$, $\beta = .21$, $p = .009$), and PCK ($B = .28$, $\beta = .32$, $p < .001$) made statistically significant positive contributions to TPACK scores. CK ($B = -.15$, $\beta =$

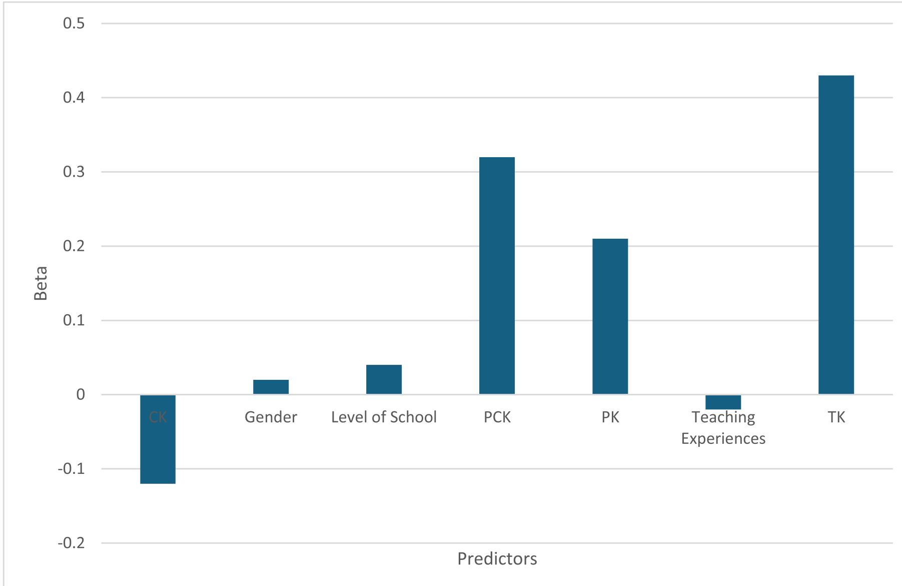

Fig. 3. The Prediction of the impact of variables on teachers' TPACK-GenAI competencies.

−.12, p = .096) was a negative but non-significant predictor. Teaching experience (B = −.04, β = −.02, p = .572), level of school (B = .11, β = .04, p = .320), and gender (B = .15, β = .02, p = .532) were also non-significant predictors. These findings suggest that teachers' TK, PK, and PCK are the strongest and most reliable predictors of their TPACK, whereas demographic factors such as teaching experience, level of school, and gender did not significantly influence TPACK levels. Therefore, hypothesis 2 is rejected.

4.3. Mediation analysis

Mediation analysis was conducted to examine whether TPK mediates the relationship between TK and overall TPACK-GenAI. This analysis was performed using ordinary least squares (OLS) regression based on the PROCESS macro (Model 4) by Hayes (2017), with a bootstrap sample of 5000 to assess the significance of the indirect effects. The results provide insights into the mechanism through which teachers' TK contributes to their broader competence in integrating GenAI in pedagogical practices. The corresponding mediation model, presented in Fig. 4, illustrates the direct and indirect pathways tested in the analysis, including the effect of TK on TPK (path a), the effect of TPK on TPACK (path b), and the direct effect of TK on TPACK (path c), along with the indirect effect computed through bootstrapping.

The mediation analysis examined the indirect effect of TK on TPACK through TPK. The findings reveal that TK significantly predicts TPK (β = 1.09, p < .001), explaining approximately 55.3% of the variance in TPK. This suggests that TK plays a crucial role in developing pedagogical strategies for integrating technology in teaching (See Table 5).

When both TK and TPK were included in the model predicting TPACK, TK had a small but statistically significant direct effect (β = .06, p = .0345), whereas TPK remained a strong and significant predictor (β = .35, p < .001). The total effect of TK on TPACK (β = .44, p < .001) was significant, and its direct effect remained significant, indicating partial mediation. The indirect effect of TK on TPACK through TPK was significant (β = .39, 95% CI [.32, .46]), confirming that 87% of the effect of TK on TPACK occurs through its influence on TPK. These findings suggest that TPK partially mediates the relationship between TK and TPACK, therefore, hypothesis 3 is accepted. TK directly contributes to TPACK to a small extent; its primary impact is exerted indirectly via its influence on TPK. Teachers' ability to integrate technological and pedagogical strategies plays a crucial role in translating technological proficiency into broader TPACK competence (see Fig. 4).

Table 4
Multiple regression analysis predicting TPACK from knowledge domains, teaching experience, school level, and gender.

Variables	R^2	ΔR^2	B	B	F	95% Cl LL	95% Cl UL
Model	.58	.57			61.35		
Constant			1.54			−.19	3.27
TK			.28	.43		.21	.34
CK			−.15	−.12		−.34	.03
PK			.15	.21		.04	.25
PCK			.28	.32		.17	.40
Teaching Experience			−.04	−.02		−.19	.11
Level of School			.11	.04		−.11	.32
Gender			.15	.02		−.31	.60

Note. B = Un-standardized Coefficient; *β= standardized coefficient CL* = confidence interval; *LL* = lower limit; *UL* = upper limit; Dependent variable is Technological pedagogy and content knowledge (TPACK).

4.4. Moderation analysis

A moderation analysis was conducted using Hayes' (2017) PROCESS macro (Model 1) in SPSS to examine whether Level of School moderates the relationship between TK and TPACK. The model included TK as the focal predictor, Level of School as the moderator, and TPACK as the outcome variable. Level of School was entered as a continuous ordinal variable coded as 1 = Kindergarten, 2 = Primary, 3 = Middle School, and 4 = High School. In the PROCESS output, the terms 'lower' and 'higher' school levels refer to the conditional values at −1 SD and +1 SD of this ordinal variable, respectively, which is the default probing method in PROCESS Model 1 for continuous moderators (Hayes, 2017). Fig. 5 illustrates the moderation effect, showing the conditional slopes of TK predicting TPACK at each level of the moderator. As indicated in the graph, the positive association between TK and TPACK is evident at both school levels, but is stronger for teachers in lower school levels compared to higher school levels.

The overall model was statistically significant, R2 = .47, p < .001R^2 = .47, p < .001R2 = .47, p < .001, indicating that TK, Level of School, and their interaction together explained approximately 47% of the variance in TPACK. As mentioned in Table 6 and Fig. 6, the main effect of TK was positive and highly significant (B = .61, p < .001), suggesting

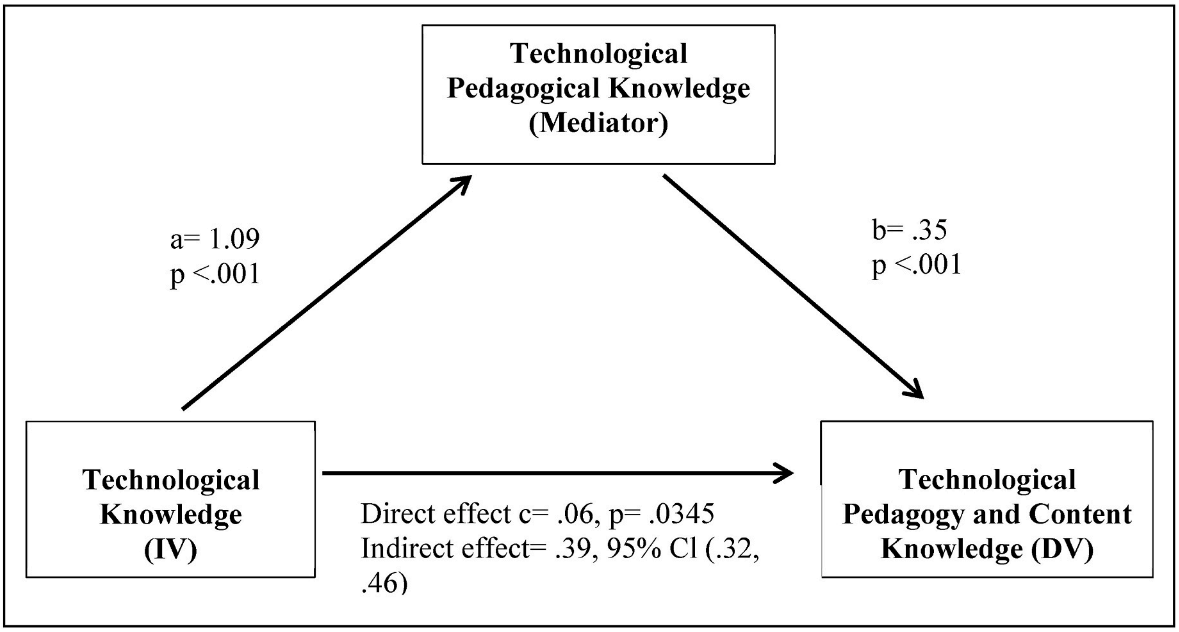

Fig. 4. The indirect effect of TK on TPACK through TPK

Table 5
Indirect effect of technological knowledge in predicting technological pedagogical content knowledge (TPACK) among teachers (N = 325).

Predictor variable	Outcome variable	*B*	*P*	*95% CI*	
				LL	UL
Total Effect		.44	<.001	.39	.49
Direct Effect					
TK	TPK	1.09	<.001	.98	1.20
TPK	TPACK	.35	<.001	.32	.39
TK	TPACK	.06	.0345	.00	.11
Indirect Effect					
TK	TPACK through TPK	.39		.32	.46

Note**.** *TK* = Technological Knowledge; *TPK* = Technological Pedagogical Knowledge; *TPACK* = Technological Pedagogical Content Knowledge; β = Unstandardized Coefficient; *CI* = Confidence Interval; *LL* = Lower Limit; *UL* = Upper Limit.

that higher TK scores are associated with higher TPACK, holding Level of School constant. Level of School also showed a significant positive effect (B = 2.82, p = .031), indicating that teachers at higher school levels tend to have higher TPACK scores when TK is held constant. The interaction term between TK and Level of School was significant (B = −.11, p = .037), suggesting that the strength of the relationship between TK and TPACK depends on the Level of School. The negative coefficient for the interaction indicates that the slope of TK predicting TPACK is weaker at higher school levels compared to lower school levels. This finding is visually supported by the moderation plot (Fig. 5), where the line for lower school-level teachers has a steeper positive slope than that for higher school-level teachers. Therefore, hypothesis 4 is accepted.

4.5. Interviews analysis

Seven in-service teachers were interviewed virtually via Zoom to gain deeper insights into their TPACK-GenAI confidence and integration of GenAI tools in classroom practices. The participants selected for the interviews had previously attended professional development training on using GenAI in education**,** while those without such training were excluded. The interview discussions focused on teachers' experiences, perceptions, and challenges in applying AI tools across the study's three domains: *AI for content creation and instructional* support, *AI for teaching and learning*, and *AI for assessment and feedback*. Participants' responses were analyzed and categorized thematically according to these domains to ensure alignment with the study objectives.

4.5.1. Q1: what type of AI do you believe has the greatest potential to revolutionize education, and why?

4.5.1.1. Perceptions of AI's potential. Participants in this study consistently emphasized that generative and supportive AI tools, for instance, ChatGPT, MagicSchool, and Gamma AI, are transforming their educational experiences. Many participants shared that these AI tools have become crucial in streamlining lesson planning, generating assessments, and creating engaging learning materials. For example, MagicSchool helps teachers to design curriculum and assessments efficiently, while Gamma AI simplifies the creation of visually engaging presentations.

Considering the significance of MagicSchool, one teacher educator remarked, *"MagicSchool helps me prepare complete lessons in minutes it's a game changer to make difference in students' learning"* (**T1**).

Other participants complimented Gamma AI for enabling students to develop visually engaging content rapidly, increasing students' engagement. ChatGPT and DeepSeek are commonly stated as reliable virtual assistants that offer prompt access to knowledge and ideas, therefore supporting both students and teachers in discovering topics more deeply.

Rather than perceiving AI as a risk, participants viewed these tools as valuable partners in the teaching-learning process.

As one educator explained it, *"Generative AI is seen as a smart assistant to educators not a replacement enhancing quality, flexibility, and fairness in education"* (**T3**).

These perspectives reflect a stronger sentiment among participants that AI increases teacher productivity, promotes personalized learning, and supports more equitable access to education indicating a promising future where human intelligence is amplified, not replaced, by AI.

4.5.1.2. Adaptive and personalized learning as the future of student-centered education. Participants highlighted that adaptive and personalized AI tools are restructuring classrooms into more inclusive, student-centered environments. Many educators shared that AI's real-time ability to measure learning styles, monitor progress, and regulate content based on students' performance has made the teaching and learning

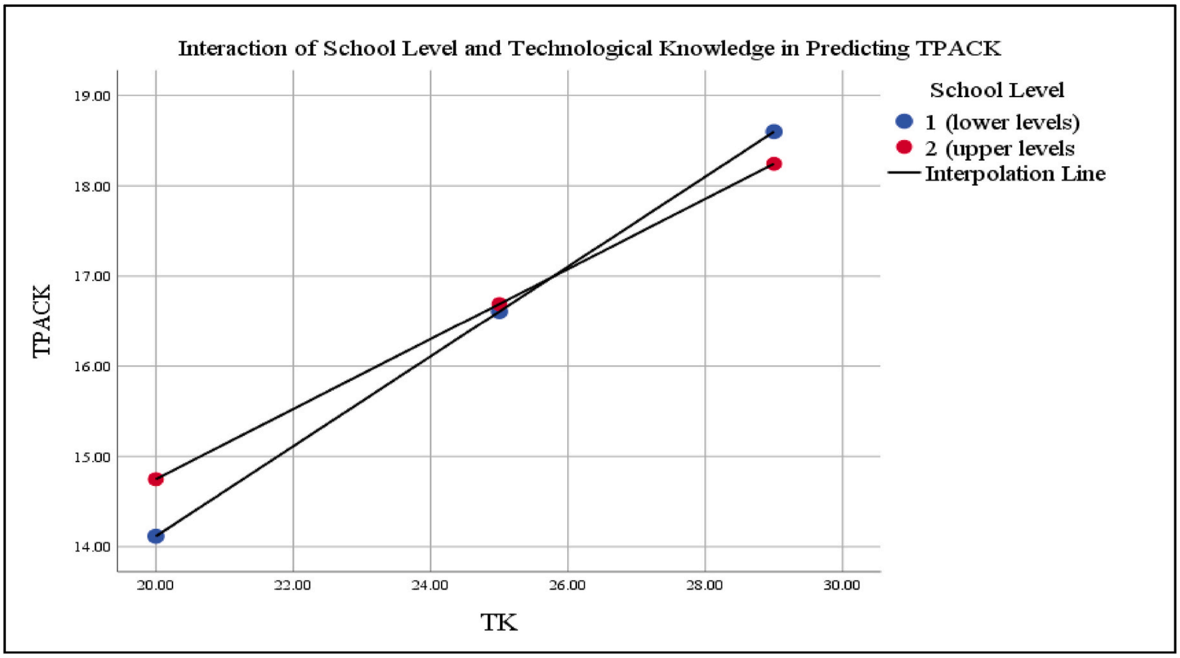

Fig. 5. School level as a moderator of the TK-TPACK relationship.

Table 6
Moderation analysis predicting TPACK from TK and level of school.

Predictors	B	SE	95% CL	
			LL	UL
Main Effects				
TK	.61***	.08	.44	.77
Level of School	2.82*	1.30	.26	5.39
Interaction Effect				
TK × Level of School	−.11*	.05	−.21	−.01
Model Summary				
Total R^2	.47***			
R^2 Change	.0072*			

Note. *B* = Un-standardized Coefficient; *S.E* = Standard Error; *CI* = Confidence Interval; *UL* = Upper Limit; *LL* = Lower Limit; outcome variable is TPACK; ***p < .001; **p < .01; *p < .05.

process more responsive and effective.

One teacher noted, *"AI identified where my students are struggling before I even had a chance to review their work and monitor their performance"* (**T4**).

Such insights allow for the development of personalized learning plans that adapt to each learner's pace and ability, ensuring no student is left behind.

As one participant described, *"Adaptive AI is complimented for its ability to analyze students' learning styles, levels, and needs in real time"* (**T2**).

This capability supports self-paced learning and targets tailored interventions that empower students to learn in those ways that suit their distinctive needs. Numerous participants also highlighted how adaptive AI promotes educational equity by supporting diverse learners, particularly those from varying academic, linguistic, and social backgrounds. By customizing teaching, adaptive technologies aid to bridge learning gaps, making quality education more accessible and impartial to all students.

4.5.2. Q2: can you describe an innovative use of AI in education that you found particularly effective, and explain what made it stand out?

4.5.2.1. AI for content creation and instructional support. Participants across the board highlighted the role of GenAI tools that have become integral to the daily planning and instructional responsibilities of teachers. This shift has brought about a paradigm change in the classroom, where teachers are no longer burdened with monotonous planning tasks. Instead, they can redirect their energy toward more significant aspects of teaching, such as student engagement, differentiated instruction, and promoting critical thinking skills. Several participants believed that these tools serve as *smart assistants* not only reducing time spent on manual work but also enhancing the creativity and adaptability of instructional materials. Gamma AI was acclaimed by the participants for creating visually engaging content that meets both educational and design standards.

Another participant signifies the importance of AI tools for content creation and highlights how AI enhances not just efficiency but also the creative quality of classroom materials. *"Gamma AI makes beautiful presentations and worksheets with just a few prompts it's like to have a design assistant in the classroom"* (**T1**).

The responses indicate that participants did not see AI as replacing educators, but rather as a supportive force that empowers teachers to be more efficient, student-centered, and innovative. The enhanced speed, design, and personalization offered by these AI tools stood out to most participants as transformative elements of modern education.

4.5.2.2. Personalized learning through adaptive AI. Another prominent theme emerging from participants' responses is the transformative role of adaptive AI in promoting personalized learning experiences. Many participants emphasized how AI tools can robustly regulate instruction based on students' individual learning styles, capabilities, and learning development. These tools examine performance data in real time to modify content, adjust difficulty levels, and provide prompt feedback permitting each learner to move at their own pace. This was perceived as particularly useful in mixed-ability classrooms, where some students may need extra time or alternative support approaches.

One participant valued this learning and expressed: *"I've seen AI tools like ClassPoint and others analyze student performance in real time and suggest follow-up activities instantly it acts like to have a co-teacher"* (**T4**).

Other participants noticed the impact of adaptive AI that stimulates a more inclusive and equitable environment, proposing targeted interventions for struggling students without singling them out.

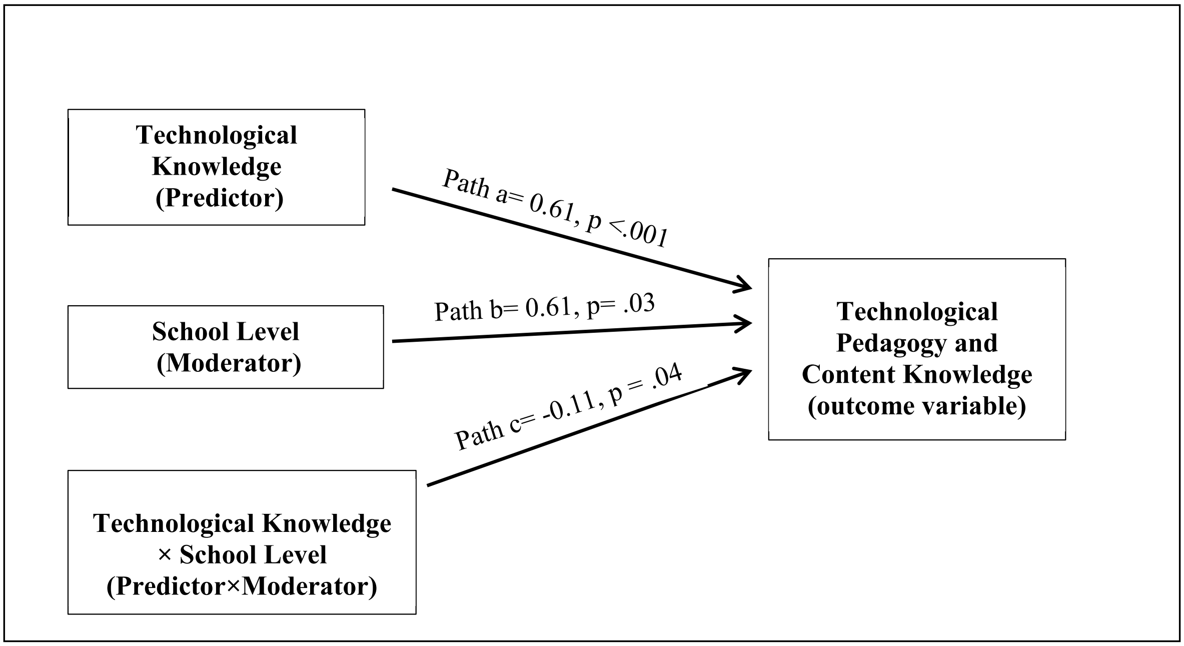

Fig. 6. The moderation effect of school level on the relationship between TK and TPACK

A participant reflected, *"AI helped one of my struggling students stay on track without feeling left out, as it adapted to her* pace*"*. (**T2**).

One of the most prominent aspects of using AI in education is the sense of empowerment it brings to both teachers and students. These tools transform AI into a personalized tutor that adapts to individual learning needs, offering support, guidance, and feedback in real time.

4.5.3. Q3: Please share a specific example of how you have used AI to enhance student learning and engagement in your classroom. What factors contributed to its success?

4.5.3.1. Classroom application. Participants responded that AI tools were successfully leveraged to produce adaptive learning environments that not only reinforced personalized instruction but also enhanced student engagement. They shared that one practical example was using *ChatGPT* to design diverse question types, such as multiple-choice, true/false, and fill-in-the-blank activities. This approach introduced a range of assessment formats that provided different learning preferences and reinforced comprehension through innovative engagement.

One teacher educator described *ChatGPT* as *"a powerful source of designing assessment strategies"* (**T1**). Another teacher said: *"ChatGPT understands the student's level and provides personalized content, giving each student a unique experience"* (**T6**).

This demonstrates the robust role of AI that tailors' educational materials to individual student needs, allowing them to progress at their own pace and in a way that suits their learning style.

In addition, teacher educators believed the role of other platforms like *Quizizz* that were used to facilitate interactive learning sessions. These tools provided immediate feedback and enabled teachers to track student progress.

Regarding the significance of immediate feedback through AI tools, as one participant stated: *"AI helps in identifying students' understanding and pinpointing areas of weakness and strength for each student and working to close the gaps"* (**T7**).

Such formative assessment not only supports differentiated instruction but also enables educators to make data-driven decisions to improve learning outcomes. Through these approaches, students remained engaged, determined, and better sustained in their academic journey.

4.5.3.2. Emotional connection and student-centered creativity through AI. Another influential theme that appeared is the creative and emotional integration of students' identities into the learning process through AI. This approach made learning not only more applicable but also emotionally meaningful for the students. One standout example was the use of students' own photos to create videos during a lesson on the topic of *Honesty*.

As one teacher educator said:

> In the lesson on the topic of honesty, I used students' photos to create videos that illustrated different aspects of honesty to promote a sense of ownership, pride, and emotional investment in the lesson (**T4**).

Similarly, participants used AI-generated visuals for motivating students by holding trophies in their hands, saying: *"I uploaded students' photos with a trophy to AI, and it gave me a photo that enhanced students' motivation, self-esteem and sense of ownership"* (**T2**).

This kind of visual recognition can significantly improve students' self-esteem and incentivize positive behavior and achievement, specifically in younger students.

In conclusion, the creative integration of AI into classroom activities such as animating children's drawings and giving them voices not only enriched the learning experience but also cultivated deeper emotional connections. By blending technology with storytelling and imagination, teachers were able to design immersive, personalized lessons that stimulated students' curiosity, encouraged active participation, and made learning more joyful and meaningful.

4.5.4. Q3: what challenges do you face in integrating AI tools into your teaching practice, and what kinds of institutional support or resources would help you use AI more effectively?

4.5.4.1. Lack of access, resources, and structure. One of the prevailing challenges is the digital divide, where many schools, especially those in low-fee or underserved areas, lack necessary technological structure to support AI integration. This comprises inadequate access to devices, stable internet, and updated tools, limiting both teachers and students. Teacher educators believed that it is challenging for students to own an electronic device due to the recurring monthly costs.

One educator expressed the following:

> I believe the main problems of unavailability of modern electrical equipment, inadequacy of information technology hardware and unavailability of consistent internet. Devices are not always available for all students. The internet, sometimes some apps are blocked by school WiFi (**T4**).

This barrier is compounded by the high cost of AI tools, lack of paid subscriptions, or dependency on free models that often restrict important features. These restrictions especially affect low-income students, widening educational inequality. One of the participants said: *"AI tools are expensive. Some programs require subscription, If I pay for these resources, that is very expensive that I cannot manage from my monthly salary"* (**T3**).

Even when the school has some access, unstable connections, old-fashioned tools, and technical problems obstruct meaningful use. Exclusive of modern, reliable infrastructure, AI-enhanced learning remains out of reach for many students. One of the educators emphasized:

> Weak internet connection and paid applications are out of reach. Internet disconnection and slow speed issues and limited resources require support to get access on desired platforms (**T5**).

4.5.4.2. Lack of training, Awareness, and pedagogical support. Another main barrier recognized is a widespread lack of training and confidence among teacher educators to utilize AI tools. Many teachers felt untrained to use AI efficiently due to inadequate professional development and uncertain pedagogical frameworks. One teacher educator expressed in the given situation: *"I have lack of familiarity with AI and have no teaching experience in AI. I know many teachers aren't fully trained in how AI works or how to use it effectively"* (**T6**).

This leads to resistance or hesitation, often stemming from an anxiety of losing control over the classroom, a lack of institutional support, or concern that AI may undermine the traditional role of the teacher.

Some teachers or administrators may fear that AI will replace their role or complicate the educational process. Many older teachers tend to rely solely on their experience and often do not favor new methods or approaches due to their lack of confidence. Moreover, there are growing concerns about students' over-reliance on AI, where learners use it as a support instead of a tool for development. This, paired with inadequate ethical guidance, can lead to reduced analytical thinking, plagiarism, and disengagement.

One teacher educator expressed their opinion on this situation:

> Students using [AI] to enable themselves instead [of] using it as a tool. AI has been found to kill the student's creativity. Artificial intelligence has turned the student into a machine of no benefit … a captive of technological operations (**T1**).

This recommends a crucial need for structured training, ethical policies, and continuing support to ensure that AI is employed in a way that enhances, rather than replaces, human learning and interaction.

4.5.4.3. Access to tools, devices, and infrastructure. Despite being motivated and provided knowledge many teachers encounter practical constraints that hamper effective implementation of AI-driven pedagogies. A major obstacle is the lack of physical resources: not enough computers, smart devices, or even established internet. In many classrooms, students still share devices, or rely on outdated technology, which makes it challenging to use AI-based platforms or tools in a significant way.

Moreover, teachers stated the need for interactive tools and educational technologies such as robots or AI-enhanced games, which can make learning more appealing particularly for younger students. Many were also entitled to dedicated technical support staff to help with installation, troubleshooting, and maintenance of technological gadgets in classrooms. One teacher educator shared their personal experience stating: *"Access to user-friendly AI tools, proper training workshops, and* technical support *would greatly help"* (**T3**).

This shows the unified nature of needs: even if the tools exist, teachers also need the training and support system to use them effectively.

Another educator said:

> Availability of computers for each child and a course to teach the children is a big deal. A direct call for equity in access ensuring every student has a device and learns how to use AI themselves, not just watch the teacher how to use it (**T4**).

Providing educational robots and games for children, iPads, and stable internet access can stimulate students' interest in learning. This response suggests using age-appropriate AI tools and playful learning platforms to teach children, along with a reminder that reliable connectivity is foundational to making any of this work.

5. Discussion

This study examined in-service teachers' TPACK-GenAI and explored factors influencing the integration of GenAI in classroom practice. The results showed moderate to high levels of self-reported competency across TPACK dimensions, with teachers reporting higher confidence in PK and CK than in AI-integrated factors. In general, teachers demonstrated a progressive orientation toward integrating GenAI into teaching and learning (Lakhe Shrestha et al., 2025). It is important to note that while professional development was initially hypothesized as a predictor, it was not included in the final regression model due to limitations in the collected data. The survey did not capture the type, duration, or quality of professional development experiences with sufficient detail to create a reliable variable for analysis. This represents a limitation of the current study and a key direction for future research, as the qualitative findings highlight the critical role of effective training.

5.1. Relationship between TPACK and GenAI

The initial findings of this study revealed that in-service teachers exhibited moderate to high confidence in PK and CK, while their confidence in integrating GenAI technologies into classroom practice (TPK, TPACK, and TPACK-COM) was relatively lower. This pattern aligns with earlier studies indicating that teachers are mostly more contented with pedagogy in relation to subject matter than with emergent technologies (Koehler & Mishra, 2010; Chai et al., 2016). The significant positive correlations among all constructs reveal that as teachers' confidence in one knowledge domain increases, associated competencies also improve. This interrelationship supports the inclusive view of the TPACK framework, highlighting that effective AI integration requires composed development across TK, PK, and CK (Mishra & Koehler, 2006). These results indicate that professional development programs should emphasize accurate, practice-based learning experiences to reinforce teachers' GenAI-related TPACK proficiencies (Zawacki-Richter et al., 2019; Zhai et al., 2021). Teachers who are more self-reliant in technology use are also more capable of designing AI-supported learning environments that improve engagement and personalization (Holmes et al., 2019). In general, these findings emphasize the need for sustained, practice-oriented professional development that holistically reinforces teachers' technological, pedagogical, and content integration for effective GenAI-supported teaching.

5.2. Factors influencing TPACK-GenAI competencies of teachers

The regression analysis revealed that TK, PK, and PCK considerably predicted teachers' general TPACK-GenAI competencies, while CK and demographic factors, for instance teaching experience, school level, and gender, did not predict significantly. These findings recommend that teachers' ability to incorporate GenAI in instruction mainly depends on their confidence and competence in conjoining technology with pedagogy and content. This finding resonates with previous research demonstrating that TPACK development is essential to be driven by teachers' energetic commitment with technology-enhanced pedagogical practices rather than by demographic variables (Prasetya et al., 2024).

Effective GenAI integration demands both technical proficiency and a deep pedagogical understanding in what way AI tools improve teaching and content delivery (Lakhe Shrestha et al., 2025). The non-significant effects of teaching experience and the school level variable suggest that even veteran teachers may face challenges in GenAI integration without targeted professional development. This focusses the need for sustained context-specific programs that enhance AI literacy and model practical applications (Huang et al., 2025). Collaborative and experiential learning environments further improve teachers' confidence in exercising AI for instruction and assessment (Guler et al., 2024). Overall, reinforcing teachers' TK, PK, and PCK through continuous professional development is fundamental for refining their TPACK-GenAI competency and effective AI-supported teaching.

In conclusion, these results show that teachers' successful integration of GenAI hinges on strengthening the interconnection between their TK, PK, and CK rather than relying on demographic factors.

5.3. TPK's mediating influence on teachers' use of TK

The mediation analysis reported that TPK significantly mediates the relationship between TK and overall TPACK, indicating teachers' ability to transform their technological expertise into effective instructional practices be contingent mainly on their pedagogical application of technology. However, TK had a small direct effect on TPACK; its indirect effect through TPK was substantially stronger, suggesting that technological expertise alone does not ascertain effective technology integration unless it is complemented by pedagogical understanding (Koehler & Mishra, 2009). This finding is in line with earlier work by Chai et al. (2016) who reported that TPK acts as a critical connection between acquiring technological skills and effectively applying them in contextually meaningful teaching scenarios. This reflects that teachers must move beyond being conscious of the way AI tools function to developing pedagogically comprehensive strategies for their classroom use (Zhai et al., 2021). The robust arbitrating role of TPK emphasizes the significance of professional development programs that not only enhance teachers' TK but also promote their pedagogical confidence in exercising GenAI for learning teaching design and assessment. Training teachers to integrate AI tools in relation to pedagogical framework, for instance inquiry-based or tailored instruction, significantly improves their TPACK competence and teaching confidence (Huang et al., 2025). TPK thus acts as the bridge between technological proficiency and pedagogical innovation, enabling teachers to use GenAI tools for creativity, differentiation, and deeper learning (Holmes et al., 2019).

5.4. Impact of school level on TPACK-GenAI competencies

The findings indicate that school level significantly moderates the relationship between teachers' TK and their overall TPACK-GenAI competencies, describing approximately half of the variance in TPACK scores. Although TK positively predicts TPACK across all school levels, the strength of this relationship weakens at higher school levels, signifying that teachers in kindergarten and primary levels benefit more from increases in TK than those in secondary schools (Akturk & Saka Ozturk, 2019). One possible explanation is that secondary school teachers frequently operate under tighter curriculum structures and accountability pressures, including standardized examinations, which can limit experimentation with new technologies such as GenAI. High-stakes testing environments often encourage "teaching to the test", reducing opportunities for innovative or exploratory instructional approaches and thereby constraining teachers' willingness to adopt emerging tools (Gibson, 2009; Margot & Kettler, 2019), as summarized in studies of technology integration barriers.

This variation indicates the employment of a diverse style of pedagogical design and technological self-sufficiency across educational levels. In early education, teachers often embrace more exploratory and interactive pedagogies, making their technological knowledge more directly transferable to teaching practices. Conversely, secondary school teachers may encounter more inflexible curricular demands and assessment constraints, restricting opportunities to integrate generative AI creatively despite having adequate TK (Scherer, Howard, & Tondeur, 2021; Voogt et al., 2015).

The moderating effect of school level focusses the contextual nature of AI integration, emphasizing that effective use of GenAI depends not only on teachers' technological ability but also on the pedagogical opportunities provided by the school environment. These findings align with contemporary research signifying those contextual and institutional factors such as grade level, curriculum structure, and digital structure shape teachers' ways of teaching to employ AI-driven tools in their instructional design (Holmes et al., 2019). Professional development opportunities differentiated by school level, concentrating on what way GenAI can enhance inquiry, creativity, and tailored learning in early grades, while supporting subject-specific and assessment-driven integration in secondary education (Hava & Babayigit, 2025). Strengthening pedagogical support and contextual relevance tend to empower teachers at all levels to translate their technological competence into meaningful GenAI-enhanced teaching practices.

5.5. Integration of qualitative findings with hypotheses

A deeper synthesis of the quantitative and qualitative findings provides a more detailed understanding of in-service teachers' TPACK-GenAI competencies. The quantitative results established that foundational knowledge domains (TK, PK, and PCK) were significant predictors of overall TPACK, while demographic factors like teaching experience and school level were not. The qualitative data enriches this finding, suggesting a possible explanation for the non-significant results. Interview participants consistently reported that a lack of access to resources, structured training, and pedagogical support was the primary barrier to effective GenAI integration, regardless of their years of experience or the level at which they taught. For instance, one teacher noted the lack of familiarity with AI and had no teaching experience with AI. This suggests that without targeted, high-quality professional development, even veteran teachers may struggle to translate their experience into GenAI-specific competencies. This aligns with research emphasizing that teacher characteristics are less influential than targeted training in developing new technological skills (Chai et al., 2016; Hava & Babayigit, 2025; Mishra & Koehler, 2006).

Furthermore, the mediation analysis showed that TPK is the critical bridge between TK and TPACK. The qualitative data powerfully illustrated this mechanism. Teachers described their 'aha' moments not when they learned the technical features of an AI tool, but when they understood how to use it for a specific pedagogical purpose, such as creating differentiated learning paths or providing personalized feedback. As one participant explained, AI can identify students' strengths and weaknesses in their performance. This statement does not just reflect TK, but a deep understanding of how technology can serve a pedagogical goal (TPK). The challenges teachers faced, such as student over-reliance on AI and ethical concerns, further highlight that effective integration is less about technical skill and more about the pedagogical wisdom to use these tools appropriately (Lakhe Shrestha et al., 2025; Prasetya et al., 2024; Zawacki-Richter et al., 2019).

Taken together, the findings from this study deliver a clear and compelling message: for teachers to effectively integrate GenAI, institutions must prioritize pedagogy over technology. The study results demonstrate that while technical skill is a necessary prerequisite, it is the pedagogical application of that skill (TPK) that ultimately drives comprehensive competence (TPACK). The challenges reported by teachers, lack of training, resources, and ethical guidance, are not technical problems; they are pedagogical and institutional ones.

Therefore, the central challenge for the field is not simply to provide teachers with more AI tools, but to cultivate the pedagogical wisdom, institutional support, and ethical frameworks required to use them effectively.

6. Implications of the study

The findings of this study offer significant implications for educational theory, practice, and policy. Theoretically, this research extends the TPACK framework by empirically validating its structure in the context of GenAI and confirming the pivotal mediating role of TPK. This suggests that for emerging technologies like GenAI, the ability to conceptualize pedagogical applications (TPK) is the indirect association through which technical skill (TK) translates into overall TPACK-GenAI competence.

For practitioners and designers of professional development, the findings demand a shift away from generic, tool-focused training. The non-significant effect of teaching experience, coupled with the qualitative data on teacher uncertainty, indicates that all teachers, regardless of seniority, require targeted support. Professional development should be context-sensitive, addressing the unique pedagogical needs of different school levels, as suggested by the moderation analysis. For example, training for primary school teachers could focus on using GenAI to foster creativity and exploratory learning, while secondary school teacher training might emphasize subject-specific applications and assessment design. Crucially, such programs must build pedagogical confidence in using AI, not just technical proficiency, aligning with recent calls for a more human-centric approach to AI pedagogy (Chiu, 2026).

Finally, for policymakers, this study offers three concrete recommendations. First, to mandate the integration of a TPACK-GenAI framework into all pre-service teacher certification programs, ensuring that new teachers enter the profession with foundational competence in AI-driven pedagogy. Second, to earmark funding for sustained, context-sensitive professional development for in-service teachers, moving beyond one-off workshops to create ongoing professional learning communities focused on pedagogical application. Third, to establish national-level ethical guidelines for the use of AI in schools, providing clear standards for data privacy, algorithmic fairness, and the responsible use of AI-generated content. These steps are essential to ensure that the integration of AI is not only innovative but also equitable and pedagogically sound (Lan et al., 2025).

7. Limitations and recommendations

This study has several limitations. The use of self-reported survey and interview data may be subject to response bias, and the sample, drawn from selected countries and institutions, may limit the generalizability of the findings. Methodologically, future research could employ longitudinal designs to track the development of teachers' TPACK-GenAI over time, perhaps in response to specific professional development interventions. Classroom observations and analysis of lesson artifacts would also provide more objective data on how GenAI is actually used in practice, moving beyond self-reported confidence.

Substantively, this study raises several important questions. Given the critical role of TPK, future research should investigate the specific pedagogical strategies most effective for integrating GenAI across different subjects and age groups. As professional development was identified as a key need in the qualitative data, studies could compare the effectiveness of different training models (e.g., peer mentoring, project-based learning, online modules) on developing teachers' TPACK-GenAI. Finally, exploring the ethical dimensions of GenAI use in the classroom, a theme that emerged in the interviews, represents a critical and underexplored area of inquiry (Allison et al., 2025; Lan et al., 2025).

8. Conclusion

This study investigated the complex interplay of factors shaping in-service teachers' confidence and integration of GenAI. The findings confirmed the interconnectedness of the TPACK domains and highlighted the foundational importance of TK, PK, and PCK in predicting teachers' overall TPACK-GenAI competence. The central contribution of this research is the empirical confirmation of TPK as the primary mediator through which teachers' technical skills are transformed into meaningful classroom practice.

Furthermore, the moderating effect of school level demonstrates that the path to effective GenAI integration is not uniform but is shaped by pedagogical context. Although teachers are cautiously optimistic about the potential of GenAI, they face significant challenges related to training, resources, and pedagogical guidance. By providing a detailed, mixed-methods account of these dynamics, this study contributes to the scholarly conversation on teacher preparedness in the age of AI and offers a clear message: empowering teachers for the future of education requires a holistic, pedagogically-focused, and context-aware approach.

CRediT authorship contribution statement

Laila Mohebi: Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization. **Areej ElSayary:** Writing – original draft, Validation, Software, Methodology, Data curation.

Consent to participate

All participants provided informed consent before participating, in accordance with the approved research protocol.

Ethical considerations

This study received Full Ethical Clearance from the Zayed University Research Ethics Committee, chaired by Dr. Fatma F. S. Said (College of Interdisciplinary Studies). Approval Reference: ZU24_087_F, granted on 19 October 2024, valid until 19 October 2026. All procedures complied with Zayed University and UAE national ethical guidelines.

Consent for publication

Not applicable – no identifiable personal information is included in this manuscript.

Data availability statement

The datasets generated and analyzed during this study are available from the corresponding author on reasonable request. Raw data are not publicly shared to maintain participant confidentiality.

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Declaration of competing interest

The authors declare no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.caeai.2026.100599.

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